

12 (a) (i)	loudspeaker/doorbell/telephone etc.	B1	[1]
(ii)	television set/audio amplifier etc.	B1	[1]
(iii)	satellite/satellite dish/mobile phone etc.	B1	[1]
(b)	e.g. lower attenuation/fewer repeaters more secure less prone to noise/interference physically smaller/less weight lower cost greater bandwidth <i>(any two sensible suggestions, 1 each)</i>	B2	[2]
(c) (i)	ratio = $25 + (62 \times 0.21)$ = 38 dB	C1 A1	[2]
(ii)	ratio / dB = $10 \lg(P_2/P_1)$ $38 = 10 \lg(P/\{9.2 \times 10^{-6}\})$ $P = 58 \text{ mW or } 5.8 \times 10^{-2} \text{ W}$ <i>(allow 1/2 for missing 10 in equation)</i>	C1 A1	[2]

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